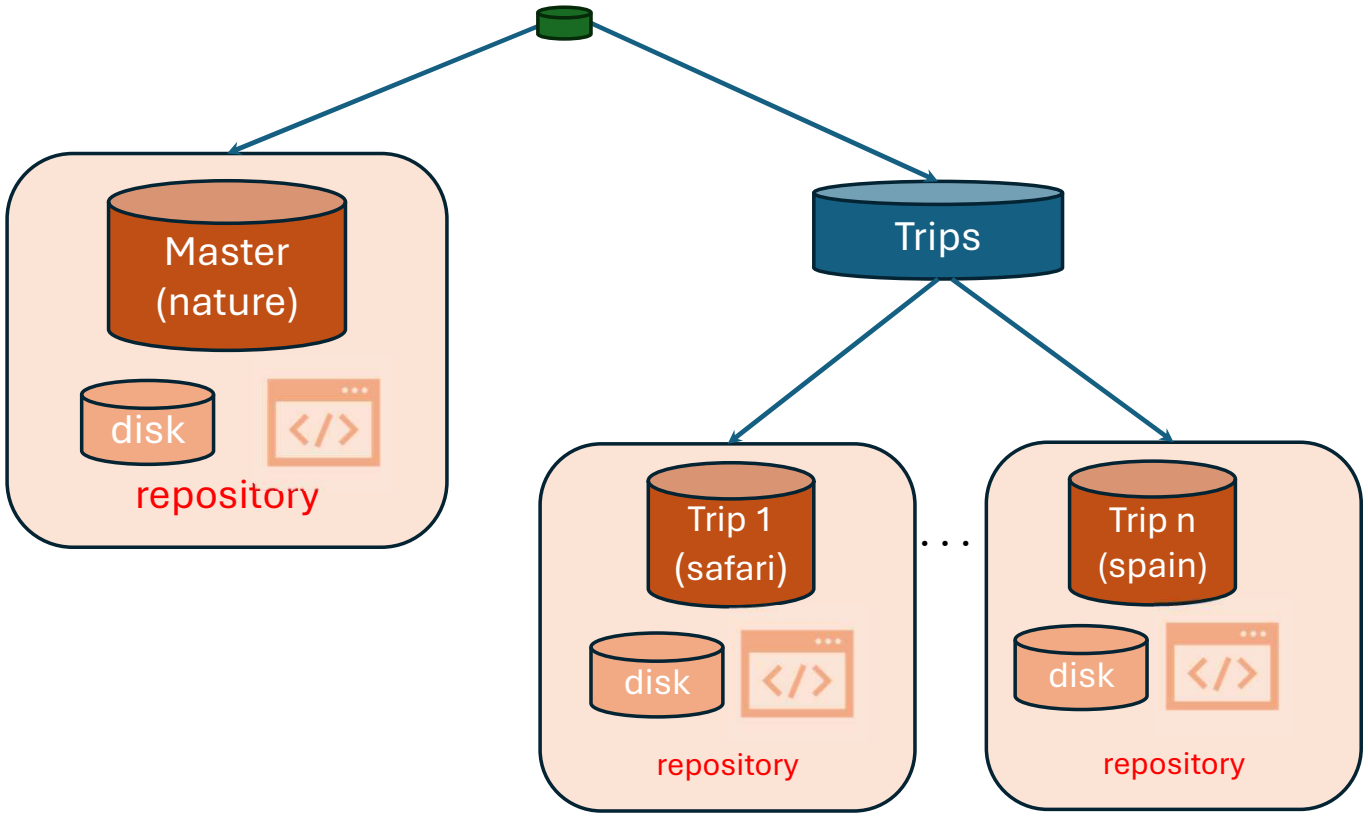
SpeciesViewer

Implementation notes

SpeciesViewer master and trip repository



Location on local servers

Master repository:
Disk: E:\dropbox\nature
URL: https:// .. /nature

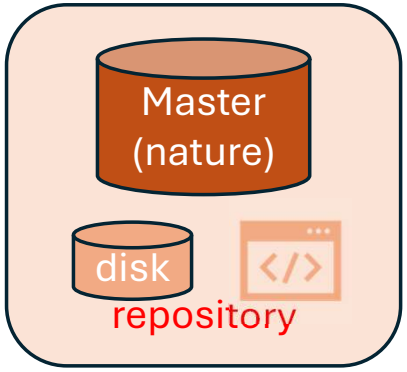
Trips have their own hierarchy:
Disk: E:\dropbox\trips\safaridb
URL: https:// .. /trips/safari

Disk: E:\dropbox\trips\spain2024
URL: https:// .. /trips/spain

The last qualifier is free-format, it is used as unique ID

Repository
(master or trip)

- **admin** has a blog administration for the repository.
- **biota** contains all species pictures, hierarchical organized. This a datastore, not aimed at publication
- **metadata** contains information all files and picture, as well as the species in our “cache” and what we actually have.
- **Web** contains a mirror of what is put on the website



admin/	• Blog_admin.csv
biota/	• animalia/ • plantae/ • fungi/
metadata/	• active/ • initialize/ • review/ (with subdirectories, see later)
Web/	• biota/ -- what is shown hierarchical on the website • biotablog/ – subdirecties every photo-blog • includes web/ – pre-installed helper html etc • summaries/ – statistics on the biota information

Files

The design is based on the concept that metadata is also be retrieved from files.

- RAWs from the camera
  - NEF or NRW from Nikon, HEIC from Apple or ORF or from Olympus
- XMP or DOP files with metadata (from Lightroom resp DxO PhotoLab)
- JPGs from the camera/iPhone, or from post-processing software

The database store in a trip is typically organized by day:

- The properties of all files are scanned with `exiftool` and saved in `data_files.csv`
- The concept of *picture* in SpeciesViewer is a set of files that are all based on a single camera click: `data_pictures.csv`. Typically this is at least a Raw and XMP file, optionally with JPGs, DOP and/or DNG (Adobe version of RAWs).
- `species.csv` contains the link between the species information an the pictures and files.
- `stats report yyyy-mm-dd.csv` summarizes the species found along with more statistics abot a directory is scanned.
- Statemachine.ini is a file to memorize which processing steps have been run on this project.

Configuration/descriptive files

SpeciesViewer adds some are .ini and .csv files in various places:

- `svconfig.ini` (variable name) – a file to describe the active repository.
- Blog entries require some text and a title for every project, which is in `blog.ini`. It is stored with the JPGs and RAWs (outside the repository)
- `admin/blog_admin.csv` – a property files that describes all blogs in a repository.
- `statemachine.ini` keeps track of progress of the process steps. It is stored in a file like `[name]/metadata/review/2024-06-26/statemachine.ini"`
- `trip.ini` – a property file that describes a trip repository, stored in the root of the trip (trips/[name]/trip.ini). Use it to give the repository a name, to provide a description, show a picture and a banner.
- `trips/Trip.csv` – like the blog_admin.csv all info from the trip.ini files.

Master and trip repositories (1)

The `svconfig.ini` defines the following directories/paths:

```
[repository]
BiotaDatebase=c:/laptop/trips/safaridb/
BiotaWebRoot=/trips/safari/
[project]
PictureDirectory=C:\CurrentPictures\2024\2024 Test Specviewer/2024-06-26
```

The same parameters work for the master and trips, but do not allow for more customization for trips, or for trips to reference existing species. This done with

```
BiotaMasterDatabase=c:/laptop/nature/
```

Based on this link, the master is available as well as (implied) the trips directory (all but last qualifier `trips/` from the `BiotaWebRoot`), and the current trip name (`safari/`)

Master and trip repositories (2)

Most of the logic is in the MyProject initialization, that reads the active svconfig.ini. `BiotaDatebase` and `BiotaWebRoot` are given.

When there a `BiotaMasterDatabase` specified, the following can be obtained:

```
ReferenceSpecies = join(BiotaMasterDatabase,
                        metadata\active\species.csv)
RepositoryBanner = trip.ini:repositorybanner
```